Lecture 2, Jan 23 - 2026.

Example: Find $(f^{-1})'(1)$ where $f(x) = 2x + \cos x$

Sln. Notice that

$$f'(x) = 2 - \sin x > 0$$

then, f is increasing and therefore f is one-to-one.

We know that

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \qquad \left\{ (f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))} \right.$$

We need $f^{-1}(1) = ?$

Notice $f^{-1}(y) = x \overset{\text{def}}{\iff} f(x) = y.$

Notice that $f(0) = 2(0) + \cos(0) = 1$ then $f^{-1}(1) = 0$.

then

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))} = \frac{1}{f'(0)} = \frac{1}{2}.$$

6.2 Exponential functions and their Derivatives.

Exponential function

In general, an exponential function is a function as.

$$f(x) = b^x$$

where b is a positive number.

If $x = n$, a positive integer, then

$$b^n = \underbrace{b \cdot b \cdot b \cdots b}_{n\text{- times}}$$

if $x=0$ then $b^0=1$ and $x=-n$ then

$$b^{-n}=\frac{1}{b^n}$$

if x is rational, $x=p/q$ where $p,q\in\mathbb{Z}$, $q>0$ then

$$b^x=b^{p/q}=\sqrt[q]{b^p}=\left(\sqrt[q]{b}\right)^p$$

Thm If $b>0$ and $b\neq 1$ then $f(x)=b^x$ is a continuous with Domain $\mathbb{R}$ and range $(0,\infty)$. In particular, $b^x>0$ for all $x\in\mathbb{R}$. If $0<b<1$, $f(x)=b^x$ is decreasing function; if $b>1$, f is increasing function. If $a,b>0$ and $x,y\in\mathbb{R}$ then:

i). $b^{x+y}=b^x b^y$

ii). $\dfrac{b^x}{b^y}=b^{x-y}$

iii). $(b^x)^y=b^{xy}$

iv). $(ab)^x=a^x b^x$

Prop. If $b>1$ then $\lim_{x\to\infty} b^x=\infty$ and $\lim_{x\to-\infty} b^x=0$

if $0<b<1$ then $\lim_{x\to\infty} b^x=0$ and $\lim_{x\to-\infty} b^x=\infty$

Example (a). Find $\lim_{x\to\infty}(2^{-x}-1)$

(b). sketch the graph of the function $y=2^{-x}-1$

Sln. a).

$$\lim_{x\to\infty}(2^{-x}-1)=\lim_{x\to\infty}2^{-x}-\lim_{x\to\infty}1$$

, b/c $\lim_{x\to\infty} c=c$

$$\lim_{x\to\infty}(f(x)\pm g(x))=\lim_{x\to\infty}f(x)\pm\lim_{x\to\infty}g(x)$$

$$=\lim_{x\to\infty}2^{-x}-1$$

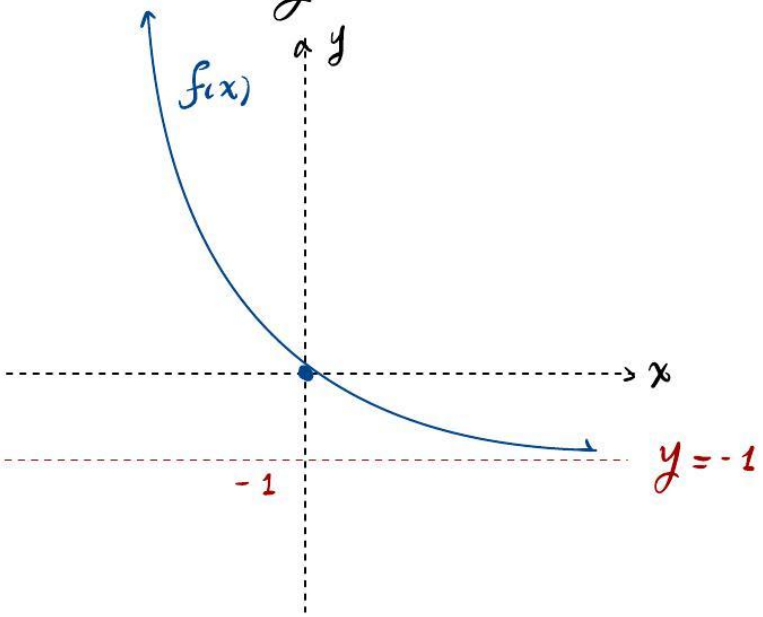
$$=\lim_{x\to\infty}\frac{1}{2^x}-1$$

$$= \lim_{x\to\infty} \left(\frac{1}{2}\right)^x - 1$$

$\left(\frac{1}{2}\right)^x \to 0$

$$= 0 - 1 = -1.$$

(b). then $y = -1$ is H.A.



Derivatives and integrals of Exponential functions.

Let $f(x) = b^x$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{b^{x+h} - b^x}{h}$$

$$= \lim_{h\to 0} \frac{b^x \cdot b^h - b^x}{h}$$

$$= \lim_{h\to 0} \frac{b^x (b^h - 1)}{h}$$

$$= b^x \lim_{h\to 0} \frac{b^h - 1}{h} = b^x \cdot f'(0)$$

Thm: If $f(x) = b^x$ then

$$f'(x) = b^x \cdot f'(0).$$

Definition of the number e.

e is the number such that $\lim_{h \to 0} \frac{e^h - 1}{h} = 1$.

Thm. $\frac{d}{dx}(e^x) = e^x$

Example. Differentiate the function $y = e^{\tan x}$.

Sln. $y' = e^{\tan x} \cdot \sec^2 x$.

prop:

$$\frac{d}{dx}(e^u) = e^u \cdot \frac{du}{dx}$$

Example. Find y' if $y = e^{-4x} \sin 5x$.

Sln:

Notice $y = f(x) \cdot g(x)$ where $f(x) = e^{-4x}$ $g(x) = \sin 5x$.

by the product rule

$$y' = f'(x)g(x) + f(x) \cdot g'(x)$$

$$\begin{cases} f'(x) = -4e^{-4x} \\ g'(x) = 5\cos 5x \end{cases}$$

$$y' = -4e^{-4x} \sin 5x + 5e^{-4x} \cos 5x$$

$$= e^{-4x}(5\cos 5x - 4\sin 5x)$$

Example. (for you). Find the absolute maximum value of the function $f(x) = xe^{-x}$.

Thm: if $\frac{d}{dx}(e^x) = e^x$ then $\int e^x dx = e^x + C$.

Example. Evaluate $\int x^2 e^{x^3} dx$.

Sln. Let $f(x) = e^{x^3}$ let $u = x^3$ $du = 3x^2 dx$.

$x^2 dx = \frac{1}{3} du$

by subst.

$$\int x^2 e^{x^3} dx = \int e^u \frac{du}{3} = \frac{1}{3}\int e^u du.$$

$$= \frac{1}{3} e^u + C.$$

$$= \frac{1}{3} e^{x^3} + C.$$

Example. Find the area under the curve $y = e^{-3x}$ from 0 to 1.

Sln. The area is

$$A = \int_0^1 e^{-3x} dx.$$

$u = -3x$

$$\int e^{-3x} dx = -\frac{1}{3}\int e^u du = -\frac{1}{3} e^{-3x} + C.$$

$-\frac{1}{3} du = dx$

$$A = \int_0^1 e^{-3x} dx = -\frac{1}{3} e^{-3x}\Big|_0^1 = -\frac{1}{3}e^{-3} + \frac{1}{3} = \frac{1}{3}(1 - e^{-3}).$$